Chapter 3: Experiment with a Single Factor: The Analysis of Variance

This chapter focuses on methods for the design and analysis of single factor experiment with an arbitrary number of k levels of the factor (or k treatments). We assume the experiment has been **completely randomized**.

ANalysis **O**f **VA**riance (**ANOVA**) is used for comparing **several population means**. In fact, we used ANOVA for testing the null hypothesis $\boldsymbol{H_0}$: $\boldsymbol{\mu_1 = \mu_2 = \mu_3 = \cdots = \mu_k}$, for some $k \geq 2$.

If the null hypothesis $\boldsymbol{H_0}$: $\boldsymbol{\mu_1 = \mu_2 = \mu_3 = \cdots = \mu_k}$is rejected, then a **detailed follow-up analysis** to decide which of the parameters differ from their hypothesized value and to estimate the size of difference can be used.

Consider the null hypothesis $\boldsymbol{H_0}$: $\boldsymbol{\mu_1 = \mu_2 = \mu_3 = \cdots = \mu_k}$, for some $k \geq 2$.

When comparing of **several populations**, the temptation is to individually test the hypotheses H_0: $\mu_i = \mu_j$ for all possible pairs of μ_i and μ_j.

The clear **weakness** of doing multiple tests is that we get many P-values, one for each test.

Question: Can we use the conventional t-tests to compare all means two at a time?

Consider the null hypothesis $\boldsymbol{H_0}$: $\boldsymbol{\mu_1 = \mu_2 = \mu_3 = \cdots = \mu_k}$, for some $k \geq 2$.

Statistical methods for dealing with **multiple comparisons** usually have two steps:

- An **overall test** to see if there is good evidence that any of the parameters differs from its hypothesized value.

- A **detailed follow-up analysis** to decide which of the parameters differ from their hypothesized value and to estimate the size of difference.

Suppose there are k populations which we would like to compare.

The $\boldsymbol{i^{th}}$ population has a **unknown mean $\boldsymbol{\mu_i}$** and **unknown standard deviation $\boldsymbol{\sigma_i}$**, where $i = 1, \ldots, k$.

We would like to see if population means are different. Therefore, we are not interested about $\boldsymbol{\sigma_1, \sigma_2, \ldots \sigma_k}$. Hence, they are called **nuisance** parameters.

Null hypothesis: H_0: $\mu_1 = \mu_2 = \mu_3 = \cdots = \mu_k$

(All population means are equal)

Alternative hypothesis H_a: H_0 is not true

(Not all population means are equal)

Null hypothesis: H_0: $\mu_1 = \mu_2 = \mu_3 = \cdots = \mu_k$

(All population means are equal)

Alternative hypothesis H_a: H_0 is not true

(Not all population means are equal)

Some students think H_a: $\mu_1 \neq \mu_2 \neq \mu_3 \neq \cdots \neq \mu_k$. **This is not true.**

Null hypothesis: H_0: $\mu_1 = \mu_2 = \mu_3 = \cdots = \mu_k$

(All population means are equal)

Alternative hypothesis H_a: H_0 is not true

(Not all population means are equal)

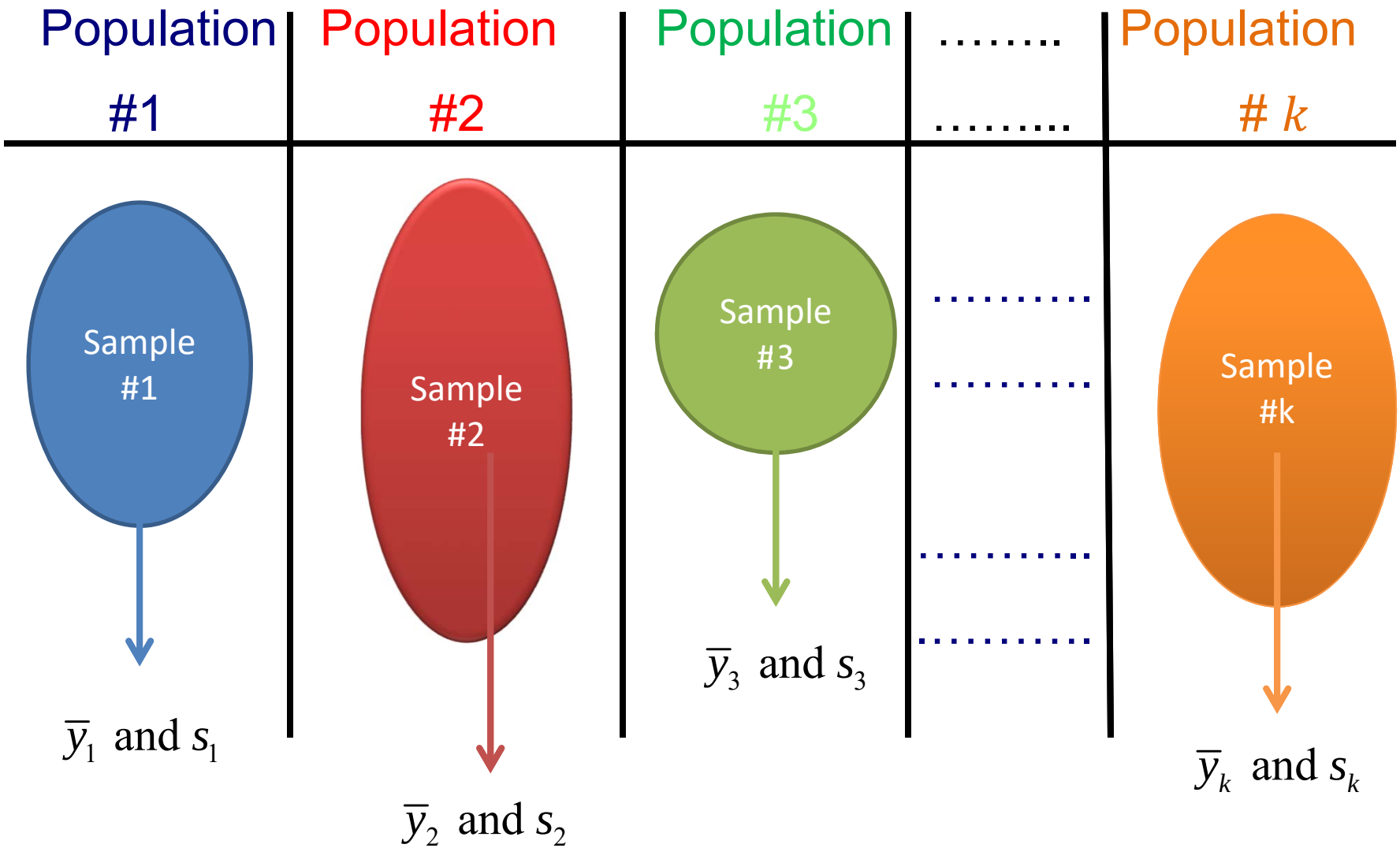
Alternative hypothesis:

H_a: **At least one** population mean is different from others.

OR

H_a: **At least two** population means are different.

To test the null hypothesis, we take **independent samples** from each population.



To test the null hypothesis, we take **independent samples** from each population. We will use above statistics to decide about null and alternative hypotheses.

Population	Sample size	Sample mean	Sample std. Dev.
1	n_1	$\bar{y}_1$	S_1
2	n_2	$\bar{y}_2$	S_2
$\vdots$	$\vdots$	$\vdots$	$\vdots$
k	n_k	$\bar{y}_k$	S_k

Total number of observations is: $\boldsymbol{n = n_1 + n_2 + \cdots + n_k}$

Necessary assumptions for **one-way** analysis of variance are:

- We have k **independent** random sample, one **Simple Random Samle** from each k populations.
- The data within each treatment group must be independent.
- The ith population has a nearly normal distribution with unknown mean μ_i where $i = 1, \ldots, k$.
- All the populations have the same standard deviation σ.

Note that: We assumed, for $j = 1,2, \ldots, n_i$:

$Y_{1j} \sim Normal(\mu_1, \sigma)$

$Y_{2j} \sim Normal(\mu_2, \sigma)$

$Y_{3j} \sim Normal(\mu_3, \sigma)$

.

.

.

$Y_{kj} \sim Normal(\mu_k, \sigma)$

If H_0: $\mu_1 = \mu_2 = \mu_3 = \cdots = \mu_k$ is true, then all populations are exactly the same. Therefore, if H_0: $\mu_1 = \mu_2 = \mu_3 = \cdots = \mu_k$ is true, then all the data appears to come from **one** giant **population**.

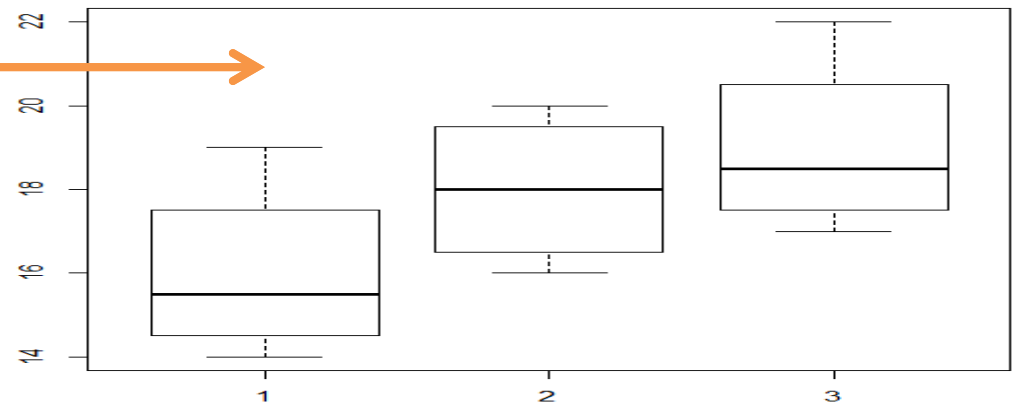
Example 1: Suppose we would like to compare three brands of gasoline. Suppose we would like to use four replicate at each brand. The following R code can be used for constructing randomization.

```
set.seed(7638)

f = factor(rep(c("Brand 1","Brand 2","Brand 3"), each=4))

fac = sample(f, 12)

eu = 1:12

plan = data.frame(Experiment =  eu, Brand = fac)

write.csv(plan, file ="Plan.csv", row.names=FALSE)
```

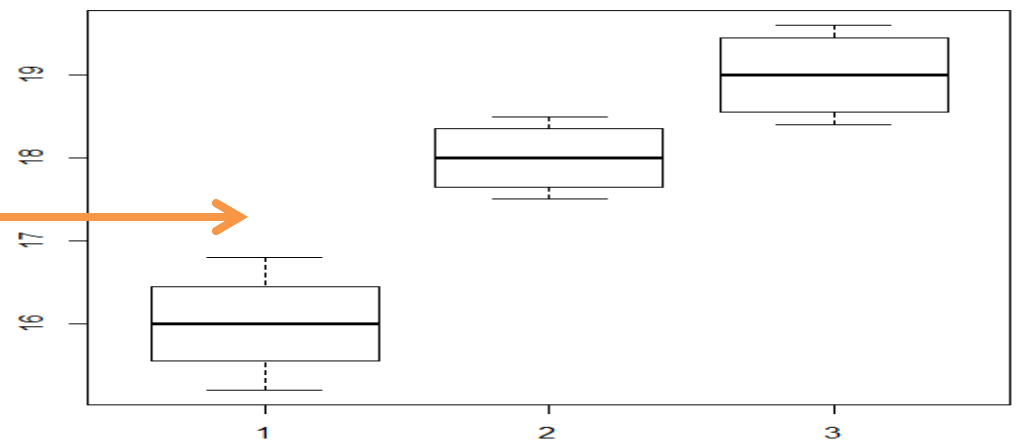
Example 1 (Cont.): Suppose we would like to compare three brands of gasoline.

Case #1

Brand 1	Brand 2	Brand 3
15	19	22
19	17	17
14	16	19
16	20	18
$\bar{y}_1 = 16$	$\bar{y}_2 = 18$	$\bar{y}_3 = 19$
$s_1 = 2.16$	$s_2 = 1.83$	$s_3 = 2.16$



Brand 1	Brand 2	Brand 3
15.2	18.5	19.6
16.1	17.5	19.3
16.8	18.2	18.4
15.9	17.8	18.7
$\bar{y}_1 = 16$	$\bar{y}_2 = 18$	$\bar{y}_3 = 19$
$s_1 = 0.66$	$s_2 = 0.44$	$s_3 = 0.55$



Example 1: Suppose we would like to compare three brands of gasoline.

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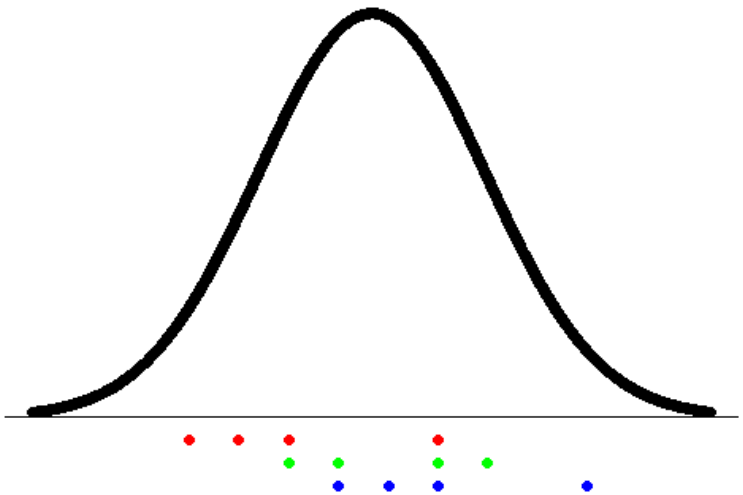
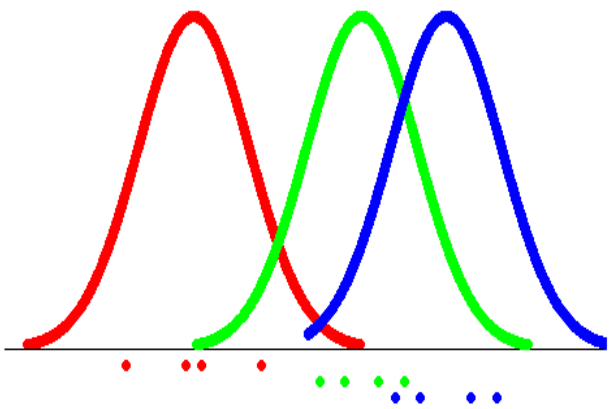


These sample averages are not the same, but this would happen even if the population means were the same.

Example 1 (Cont.): Suppose we would like to compare three brands of gasoline.

Case # 2

Brand 1	Brand 2	Brand 3
15.2	18.5	19.6
16.1	17.5	19.3
16.8	18.2	18.4
15.9	17.8	18.7
$\bar{y}_1 = 16$	$\bar{y}_2 = 18$	$\bar{y}_3 = 19$
$s_1 = 0.66$	$s_2 = 0.44$	$s_3 = 0.55$

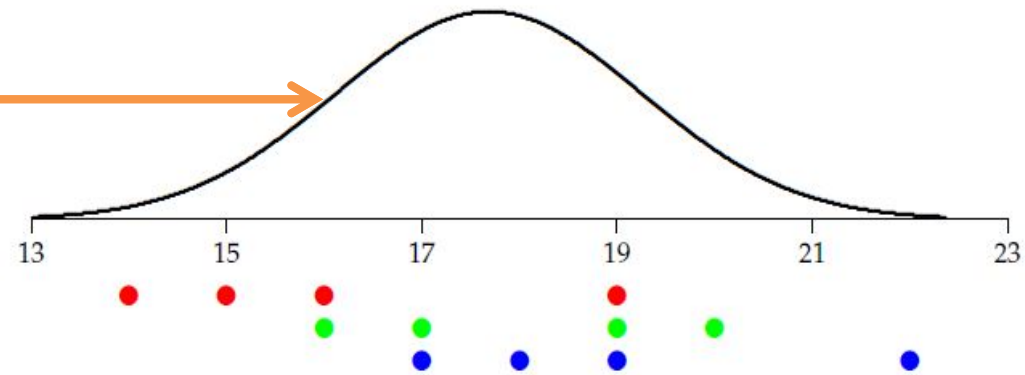


These sample averages are not the same and the variation between these sample means could be due to the variation within the populations.

Example 1 (Cont.): Suppose we would like to compare three brands of gasoline.

Case #1

Brand 1	Brand 2	Brand 3
15	19	22
19	17	17
14	16	19
16	20	18
$\bar{y}_1 = 16$	$\bar{y}_2 = 18$	$\bar{y}_3 = 19$
$s_1 = 2.16$	$s_2 = 1.83$	$s_3 = 2.16$



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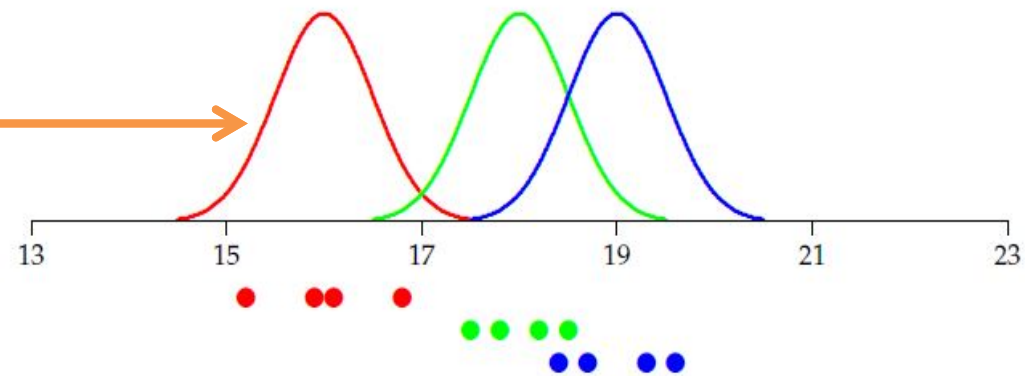
$\bar{y}_1 = 16$	$\bar{y}_2 = 18$	$\bar{y}_3 = 19$
$s_1 = 0.66$	$s_2 = 0.44$	$s_3 = 0.55$



Consider a ***model*** as a structure or set of rules that defines how the population parameters are related to each other.

For example, the *model* for the **null hypothesis** in the ANOVA says that all **means are equal**.

On the other hand, the *model* under **the alternative hypothesis** in the ANOVA implies that **the population means are not all equal**.

Therefore, the overall concept of ANOVA involves a **comparison of two competing models** and we want to test **which model is better**.

The two models can be classified as a ***reduced model*** (H_0) and ***full model*** (H_1).

Full Model: A general model that is found to adequately describe the data.

Reduced Model: It is a special case of full Model obtained by **imposing some restrictions**.

For each observation we define its **residual** as:

Residual = Observed Value – Estimated Value

Example 1: Suppose we would like to compare three brands of gasoline.

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15	19	22
19	17	17
14	16	19
16	20	18
$\bar{y}_1 = 16$	$\bar{y}_2 = 18$	$\bar{y}_3 = 19$
$s_1 = 2.16$	$s_2 = 1.83$	$s_3 = 2.16$

H_0: $\mu_1 = \mu_2 = \mu_3$ Reduced model: One Mean Model

H_A: μ_1, μ_2, μ_3 Full model: Three Means Model

Observed Value

Estimated Value for Full Model

Predicted Value

Brand	y_{ij}	$\bar{y}_{i.}$	$e = y_{ij} - \bar{y}_{i.}$	e^2
1	15	16	$15 - 16 = -1$	$(-1)^2 = 1$
1	19	16	$19 - 16 = 3$	$(3)^2 = 9$
1	14	16	$14 - 16 = -2$	$(-2)^2 = 4$
1	16	16	$16 - 16 = 0$	$(0)^2 = 0$
2	19	18	$19 - 18 = 1$	$(1)^2 = 1$
2	17	18	$17 - 18 = -1$	$(-1)^2 = 1$
2	16	18	$16 - 18 = -2$	$(-2)^2 = 4$
2	20	18	$20 - 18 = 2$	$(2)^2 = 4$
3	22	19	$22 - 19 = 3$	$(3)^2 = 9$
3	17	19	$17 - 19 = -2$	$(-2)^2 = 4$
3	19	19	$19 - 19 = 0$	$(0)^2 = 0$
3	18	19	$18 - 19 = -1$	$(-1)^2 = 1$
Total			0	38

$$SS_{\text{Residual}}(\text{Full}) = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{i.})^2 = 38$$

Observed Value

Estimated Value for Reduced Model

Predicted Value

Brand	y_{ij}	$\bar{y}_{..}$	$e = y_{ij} - \bar{y}_{..}$	e^2
1	15	17.67	$15 - 17.67 = -2.67$	7.1289
1	19	17.67	$19 - 17.67 = 1.33$	1.7689
1	14	17.67	$14 - 17.67 = -3.67$	13.4689
1	16	17.67	$16 - 17.67 = -1.67$	2.7889
2	19	17.67	$19 - 17.67 = 1.33$	1.7689
2	17	17.67	$17 - 17.67 = -0.67$	0.4489
2	16	17.67	$16 - 17.67 = -1.67$	2.7889
2	20	17.67	$20 - 17.67 = 2.33$	5.4289
3	22	17.67	$22 - 17.67 = 4.33$	18.7489
3	17	17.67	$17 - 17.67 = -0.67$	0.4489
3	19	17.67	$19 - 17.67 = 1.33$	1.7689
3	18	17.67	$18 - 17.67 = 0.33$	0.1089
Total			0.0	56.6668

$$SS_{\text{Residual}}(\text{Reduced}) = \sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_{..})^2 = 56.6668$$

Then, **the sum of squared residuals** is found by finding the residual associated with each observation, squaring each residual, and then summing them up. The sum of squared residual is calculated for each model (and). Denote:

$SS_{Residual}(Reduced)$ = Sum of squared residuals for the reduced model

$SS_{Residual}(Full)$ = Sum of squared residuals for the full model

Extra-sum-of-squared = $SS_{Residual}(Reduced) - SS_{Residual}(Full)$

The more complicated model (full model) will always have smaller sum of squared residuals.

If the **null hypothesis is true**, then the sum of squared residuals should be about the **same** for the two models.

If **the null hypothesis is incorrect**, the sum of squared residuals for the full model should be **much smaller**.

The **extra sum of squares** will determine if this **reduction is large enough** to justify the more complicated model (adding more parameters).

$$SS_{\text{Residual}}(\text{Reduced}) = \sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_{..})^2 = 56.6668$$

$$SS_{\text{Residual}}(\text{Full}) = \sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_{i.})^2 = 38 \Rightarrow \text{Extra SS} = 18.6668$$

There is a **degrees of freedom** associated with each model.

Denote:

$df_{Residual}(Reduced)$ = Residuals degrees of freedom for the reduced model

$df_{Residual}(Full)$ = Residuals degrees of freedom for the full model

Extra degrees of freedom is defined by:

$$\textbf{Extra df} = df_{Residual}(Reduced) - df_{Residual}(Full)$$

Note: The extra-degrees-of-freedom is also the number of **additional parameters** in the full mode versus the reduced model.

The **Test statistic** for testing two models is:

$$TS = \frac{Extra\ SS\ /\ Extra\ df}{MS_{Resdual}(Full)}$$

$$= \frac{SS_{Residual}(Reuced) - SS(Full)/[df_{Residual}(Reuced) - df(Full)]}{SS_{Residual}(Full)\ /\ df_{Residual}(Full)}$$

- The test statistic has an **F-distribution**, if null hypothesis is true.
- The **numerator degrees-of-freedom** for the F-distribution is **the extra degrees of freedom**.
- The **residuals degrees-of-freedom for full model** is the **denominator degrees of freedom** for the F-distribution.

Example 1:

$$SS_{\text{Residual}}(\text{Reduced}) = \sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_{..})^2 = 56.6668$$

$$SS_{\text{Residual}}(\text{Full}) = \sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_{i.})^2 = 38 \Rightarrow \text{Extra SS} = 18.6668$$

$$df_{\text{Residual}}(\text{Reduced}) = n - 1 = 12 - 1 = 11$$

$$df_{\text{Residual}}(\text{Full}) = n - 3 = 12 - 3 = 9 \Rightarrow \text{Extra df} = 11 - 9 = 2$$

The value of test statistic is: $TS = \dfrac{18.6668/2}{38/9} = 2.21$

The p-value is: $p - value = P(F_9^2 > 2.21)$

df_1

df_2

		1	2	3	4	5	6	7	8	9	10
8	0.25	1.54	1.66	1.67	1.66	1.66	1.65	1.64	1.64	1.63	1.63
8	0.1	3.46	3.11	2.92	2.81	2.73	2.67	2.62	2.59	2.56	2.54
8	0.05	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35
8	0.025	7.57	6.06	5.42	5.05	4.82	4.65	4.53	4.43	4.36	4.30
8	0.01	11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91	5.81
8	0.005	14.69	11.04	9.60	8.81	8.30	7.95	7.69	7.50	7.34	7.21
8	0.001	25.41	18.49	15.83	14.39	13.48	12.86	12.40	12.05	11.77	11.54
		1	2	3	4	5	6	7	8	9	10
9	0.25	1.51	1.62	1.63	1.63	1.62	1.61	1.60	1.60	1.59	1.59
9	0.1	3.36	3.01	2.81	2.69	2.61	2.55	2.51	2.47	2.44	2.42
9	0.05	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14
9	0.025	7.21	5.71	5.08	4.72	4.48	4.32	4.20	4.10	4.03	3.96
9	0.01	10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35	5.26
9	0.005	13.61	10.11	8.72	7.96	7.47	7.13	6.88	6.69	6.54	6.42
9	0.001	22.86	16.39	13.90	12.56	11.71	11.13	10.70	10.37	10.11	9.89

Define:

$$SS(\text{Treatments}) = SS(\text{Between Groups}) = \sum_{i=1}^{k}\sum_{j=1}^{n_i}(\bar{y}_{i.} - \bar{y}_{..})^2$$

$$df(\text{Treatments}) = k - 1$$

$$SS(Error) = SS(Within\ \text{Groups}) = \sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_{i.})^2$$

$$df(Error) = n - k$$

$$SS(Total) == \sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_{..})^2, df(\text{Total}) = n - 1$$

It can be shown that:

$$SS(\text{Treatments}) + SS(Error) = SS(\text{Total})$$

$$df(\text{Treatments}) + df(Error) = df(\text{Total})$$

The **mean square between samples** and the **mean square within samples** are defined by:

$$MS(\text{Treatments}) = MS(\text{Between Groups}) = \frac{SS(\text{Treatments})}{df(\text{Treatments})}$$

$$MS(Error) = MS(\text{Within Groups}) = \frac{SS(Error)}{df(Error)}$$

Therefore the test statistic is: $$F-test = \frac{MS(\text{Treatments})}{MS(Error)}$$

In addition,

$$\text{MS(Error)} = \text{MS(Within Groups)}$$
$$= \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2 + \cdots + (n_k-1)s_k^2}{n-k} = s_{pooled}^2 = \boldsymbol{\hat{\sigma}^2}$$

Therefore the **estimate** of **common standard deviation σ** is

$$\boldsymbol{\hat{\sigma} = \sqrt{MS(Error)}}.$$

All the calculations are usually summarized in the following table called **ANOVA table**.

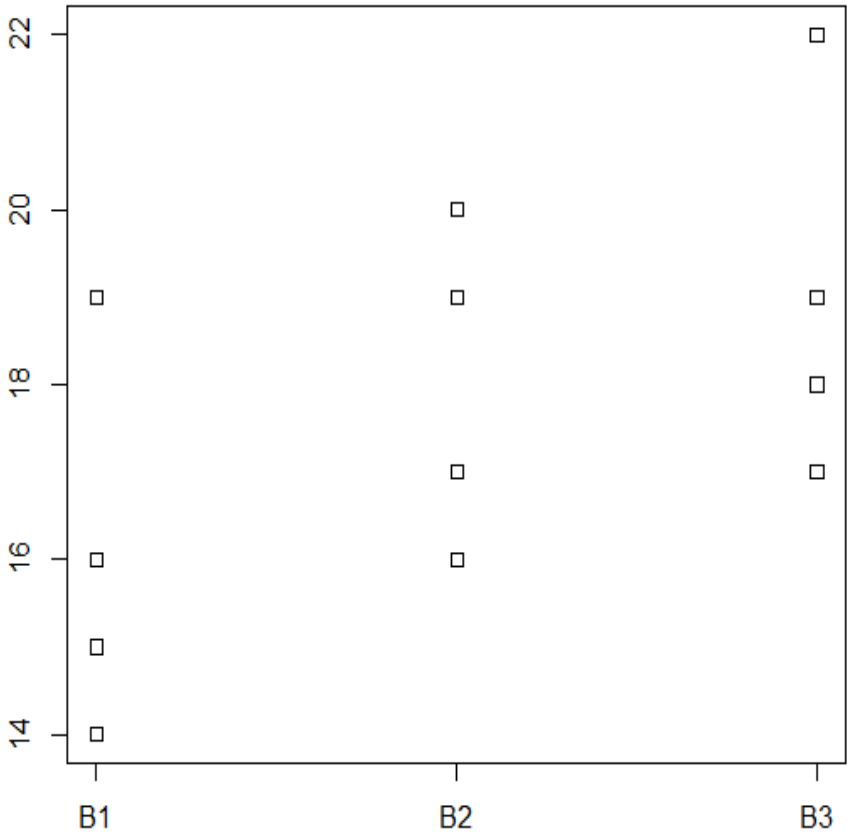
Source	df	SS	MS	F	P-value
Treatments	$k-1$	SSTr	$\mathrm{MSTr} = \frac{\mathrm{SSTr}}{\mathrm{k}-1}$	$\mathrm{F} = \frac{\mathrm{MSTr}}{\mathrm{MSE}}$	$\mathrm{P}(\mathrm{F}_{k-1,\mathrm{n}-k} > F)$
Error	$\mathrm{n}-k$	SSE	$\mathrm{MSE} = \frac{\mathrm{SSE}}{\mathrm{n}-\mathrm{k}}$		
Total	$\mathrm{n}-1$	SST			

For example 1, the ANOVA table is:

Source	df	SS	MS	F	P-value
Treatments	2	18.6668	9.333	2.21	0.1656
Error	9	38.0000	4.222		
Total	11	56.6668			

R Codes to Draw Scatter-Plot of Data and Get the Averages:

```
>B1 = c(15,19,14,16)
>B2 = c(19,17,16,20)
>B3 = c(22,17,19,18)

>data = rbind(B1,B2,B3)

>stripchart(data.frame(t(data)),
vertical = TRUE)

>ave=apply(data, 1, mean)
```

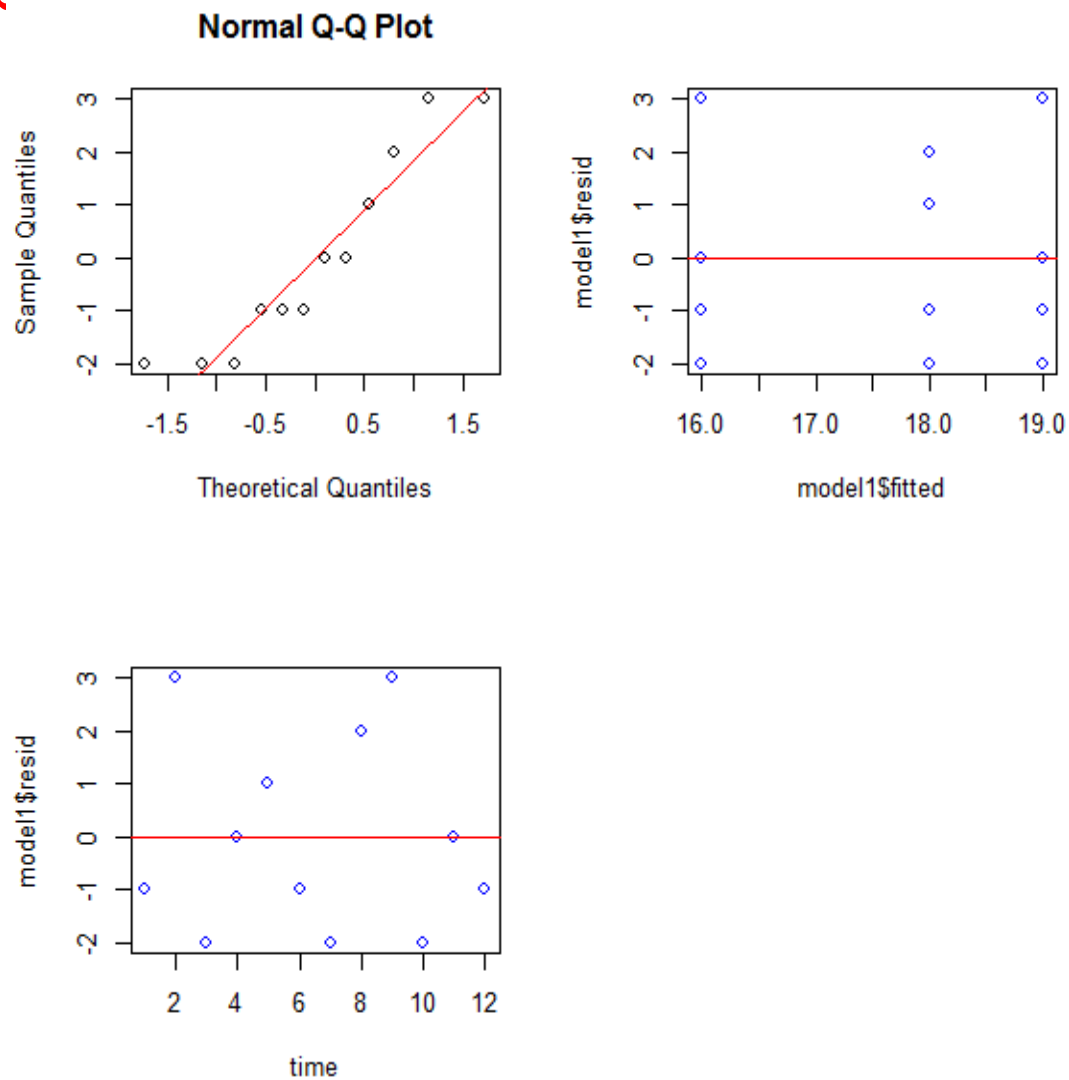


R Codes to Perform ANOVA:

```
>y = c(B1,B2,B3)

>f = factor(rep(c("Brand 1","Brand 2","Brand 3"), each=4))

> model1 = lm(y~f)
> anova(model1)

Analysis of Variance Table

Response: y
          Df Sum Sq   Mean Sq   F value   Pr(>F)
f          2  18.667   9.3333   2.2105    0.1656
Residuals  9  38.000   4.2222
```

R Codes to Check Assump

```
>windows()
>par(mfrow=c(2,2))

>qqnorm(model1$resid)
>qqline(model1$resid,
   col=2)

>plot(model1$fitted,
model1$resid, col="blue")
>abline(h=0, col="red")

>time = 1:12
>plot(time, model1$resid, col="blue")
>abline(h=0, col="red")
```



Bartlett's Test To Check Equal Variance Assumption:

Bartlett's test can be used to test the $H_0: \sigma_1 = \sigma_2 = \cdots = \sigma_k$ versus alternative that at least two variances are different. Bartlett's test statistic is:

$$TS = \frac{(n-k)Ln(s_p^2) - \sum_{i=1}^{k}(n_i - 1)Ln(s_i^2)}{1 + \frac{1}{3(k-1)}\left(\sum_{i=1}^{k}\frac{1}{n_i - 1} - \frac{1}{n-k}\right)}, \text{ where } s_p^2 = MS(\text{Error}) = \frac{\sum_{i=1}^{k}(n_i - 1)s_i^2}{n-k}$$

It can be shown that, if the null hypothesis is true, the test statistic has a chi-square distribution with $df = k - 1$. Therefore, the null hypothesis is rejected for large value of test statistic.

Note: Bartlett's test is sensitive to normality.

Bartlett's Test to Check Equal Variance Assumption:

$$s_p^2 = MS(\text{Error}) = \frac{\sum_{i=1}^{k}(n_i - 1)s_i^2}{n-k}$$

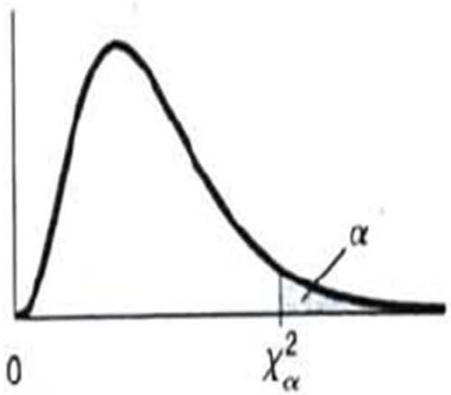
$$= \frac{(4-1)2.16^2 + (4-1)1.83^2 + (4-1)2.16^2}{12-3} = 4.22$$

$$TS = \frac{(n-k)Ln(s_p^2) - \sum_{i=1}^{k}(n_i - 1)Ln(s_i^2)}{1 + \frac{1}{3(k-1)}\left(\sum_{i=1}^{k}\frac{1}{n_i - 1} - \frac{1}{n-k}\right)}$$

$$= \frac{(12-3)Ln(4.22) - \left(3Ln(2.16^2) + 3Ln(1.83^2) + 3Ln(2.16^2)\right)}{1 + \frac{1}{3(3-1)}\left[\left(\frac{1}{4-1} + \frac{1}{4-1} + \frac{1}{4-1}\right) - \frac{1}{12-3}\right]} \cong 0.09$$

Table X

Values of χ^2_α



Right-tail probability	0.10	0.05	0.025	0.01	0.005
df					
1	2.706	3.841	5.024	6.635	7.879
2	4.605	5.991	7.378	9.210	10.597
3	6.251	7.815	9.348	11.345	12.838
4	7.779	9.488	11.143	13.277	14.860
5	9.236	11.070	12.833	15.086	16.750
6	10.645	12.592	14.449	16.812	18.548
7	12.017	14.067	16.013	18.475	20.278
8	13.362	15.507	17.535	20.090	21.955
9	14.684	16.919	19.023	21.666	23.589
10	15.987	18.307	20.483	23.209	25.188
11	17.275	19.675	21.920	24.725	26.757
12	18.549	21.026	23.337	26.217	28.300
13	19.812	22.362	24.736	27.688	29.819
14	21.064	23.685	26.119	29.141	31.319
15	22.307	24.996	27.488	30.578	32.801
16	23.542	26.296	28.845	32.000	34.267
17	24.769	27.587	30.191	33.409	35.718

R Codes to Perform Bartlett's Test :

```
>model1 = lm(y~f)

>bartlett.test(model1$resid, f)

Bartlett test of homogeneity of variances

data:  model1$resid and f
Bartlett's K-squared = 0.0946, df = 2, p-value = 0.9538
```

Levene's's Test To Check Equal Variance Assumption:

A test which is more robust against non-normality is 'Levene's test'.

Equality of the variances is indicated by equality of the group means of the absolute deviations from the group medians, which is tested by the usual F-test.

To do the test:

- Find $m_i =$ Median of sample $i = 1,2, \ldots k$.
- Calculate $d_{ij} = |y_{ij} - m_i|$ absolute deviations from the group medians.
- Perform ANOVA F-test for testing equality of mean of d_{ij}.

R Codes to Perform Levene's Test :

```
>data = rbind(B1,B2,B3)
>m=apply(data, 1, median)
>y1=c(data)
>diff <- abs(y1-rep(m, times=4))
> f1 = factor(rep(c("Brand 1","Brand 2","Brand 3"), times=4))
>model.d <- lm(diff~f1)
>anova(model.d)
```

Analysis of Variance Table

Response: diff

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
f1	2	0	0.0000	0	1
Residuals	9	13	1.4444		

You may use leveneTest() funcytion in “car” package.

R Codes to Perform Levene’s Test :

```
> install.packages("car")   #You need to install a package once
> library(car)
> y = c(B1,B2,B3)
> f = factor(rep(c("Brand 1","Brand 2","Brand 3"), each=4))
> leveneTest(y~f)

Levene's Test for Homogeneity of Variance (center = median)
        Df     F value     Pr(>F)
group   2        0           1
        9
```

If assumptions seem to be **violated** we might:

- try a **transformation** of the y_{ij} (such as $\sqrt{y_{ij}}$ or Lny_{ij}) hoping that the assumptions are valid for transformed values.

- apply a **nonparametric** procedure. In the latest case we work with the **ranks** $R_{ij}$of the y_{ij} rather than their numerical values (so this is an-other kind of transformation - the rank transformation).

Nonparametric tests are more **robust**.

If assumptions seem to be **violated** we might:

```
>y
 [1] 15 19 14 16 19 17 16 20 22 17 19 18

>R = rank(y)

>R
 [1]  2.0  9.0  1.0  3.5  9.0  5.5  3.5 11.0 12.0  5.5  9.0  7.0

>model1.rank = lm(R~f)
>anova(model1.rank)
Analysis of Variance Table
Response: R
          Df    Sum Sq Mean Sq   F value   Pr(>F)
f          2    43.875  21.938     2.054   0.1842
Residuals  9    96.125  10.681
```

Note that, apart from ties $SS_{Total}$is just $n-1$times the variance of the ranks $1,2,\ldots,n$.

- In Rank test, SS_{Total} is not random and can be calculated without knowing the data.

- Since $SS_{Eror} = SS_{Total} - SS_{Treatments}$,the only randomness is in $SS_{Treatments}$.

- There is a chi-square approximation to the distribution of $SS_{Treatments}$.

This is used in the **Kruskal-Wallis** test which typically gives results very similar to the rank transformation followed by the F-test.

R Code to Perform Kruskal-Wallis Test :

```
>kruskal.test(y,f)

        Kruskal-Wallis rank sum test

data:  y and f
Kruskal-Wallis chi-squared = 3.4473, df = 2, p-value = 0.1784
```

Example 2: The following data are the weight gains, measured in pounds, of babies from birth to six months of age. All babies in the sample weighed between seven and eight pounds at birth. One group of babies was breast-fed and the other group was fed a specific formula.

Breast -fed	10 7 8 6 8 9 9
Formula - fed	8 9 10 8 7 6

Is there evidence to suggest that the weight gains are different between the two groups of babies?

Example 2 (Cont.):

```
> t.test(y1,y2, var.equal =T)

        Two Sample t-test

data:  y1 and y2
t = 0.1865, df = 11, p-value = 0.8555
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -1.543289  1.829003
sample estimates:
mean of x mean of y
 8.142857  8.000000
```

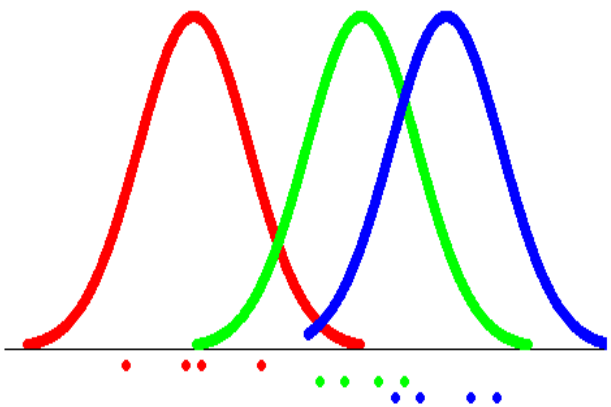
Example 2 (Cont.):

```
> y1=c(10,7,8,6,8,9,9)
> y2=c(8,9,10,8,7,6)
> y12 =c(y1,y2)
> f12 = factor(c(rep("B",7), rep("F", 6)))
 >m = lm(y12~f12)

>anova(m)
Analysis of Variance Table

Response: y12
          Df  Sum Sq  Mean Sq  F value  Pr(>F)
f12        1  0.0659  0.06593  0.0348   0.8555
Residuals 11 20.8571  1.89610
```

Example 1 (Cont.): Suppose we would like to compare three brands of gasoline.

Case # 2

Brand 1	Brand 2	Brand 3
15.2	18.5	19.6
16.1	17.5	19.3
16.8	18.2	18.4
15.9	17.8	18.7
$\bar{y}_1 = 16$	$\bar{y}_2 = 18$	$\bar{y}_3 = 19$
$s_1 = 0.66$	$s_2 = 0.44$	$s_3 = 0.55$



Analysis of Variance Table

Response: y

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
f	2	18.667	9.3333	30.216	0.0001016 ***
Residuals	9	2.780	0.3089		

Example 1 (Cont.): Suppose we would like to compare three brands of gasoline.

- Small p-value in the ANOVA tables for case 2 indicates that treatment effects are **significant**.
- Both test of equality of variance are **not** significant.
- Q-Q-plot and other plots do **not** indicate any cause for alarm.

We can now go on to answer our second question - which of the treatment means are significantly different?

Can we say with any confidence that a particular one is largest?

This is a problem of '**multiple comparisons**' - comparing several means with each other.

A confidence interval for **one** difference $\mu_i - \mu_j = \tau_i - \tau_j$, $i \neq j$, is

$$\bar{y}_{i.} - \bar{y}_{j.} \pm t_{\frac{\alpha}{2},n-k}\sqrt{MS_E}\sqrt{\frac{1}{n_i} + \frac{1}{n_j}}$$

In general, we are interested in **contrast** of treatment means $\gamma = c_1\mu_1 + c_2\mu_2 + \cdots + c_k\mu_k$, where $c_1 + c_2 + \cdots + c_k = 0$.

A confidence for γ is $\hat{\gamma} \pm t_{\frac{\alpha}{2},n-k} S.E.(\hat{\gamma})$ where

$$\hat{\gamma} = c_1\bar{y}_{1.} + c_2\bar{y}_{2.} + \cdots + c_k\bar{y}_{k.}$$

$$S.E.(\hat{\gamma}) = \sqrt{MS_E}\sqrt{\frac{c_1^2}{n_1} + \frac{c_2^2}{n_2} + \cdots + \frac{c_2^2}{n_2}}$$

Fisher Least Significant Difference (LSD) Method: A confidence interval for **one** difference $\mu_i - \mu_j = \tau_i - \tau_j,\ i \neq j,$ is

$$\bar{y}_{i.} - \bar{y}_{j.} \pm t_{\frac{\alpha}{2},n-k}\sqrt{MS_E}\sqrt{\frac{1}{n_i} + \frac{1}{n_j}}$$

Assuming a two-sided alternative, the pair of means μ_i and μ_j would be declared significantly different if

$$|\bar{y}_{i.} - \bar{y}_{j.}| > t_{\frac{\alpha}{2},n-k}\sqrt{MS_E}\sqrt{\frac{1}{n_i} + \frac{1}{n_j}}$$

The quantity $\mathrm{LSD} = t_{\frac{\alpha}{2},n-k}\sqrt{MS_E}\sqrt{\frac{1}{n_i} + \frac{1}{n_j}}$ is called **least significant difference**.

The Fisher LSD method for comparing all pairs of means control the error rate α for each individual pairwise comparisons but does not control the experiment-wise or family-wise error rate.

A **single** confidence interval for γ has the following property:

$$\boldsymbol{P}\left(\left|\frac{\widehat{\boldsymbol{\gamma}}-\boldsymbol{\gamma}}{\boldsymbol{S.E.}(\widehat{\boldsymbol{\gamma}})}\right| \leq \boldsymbol{t}_{\frac{\boldsymbol{\alpha}}{\boldsymbol{2}},\boldsymbol{n}-\boldsymbol{k}}\right) = \boldsymbol{1}-\boldsymbol{\alpha}.$$

Suppose we would like to construct **simultaneous** (or **experiment-wise** or **family-wise**)confidence intervals for $\hat{\gamma}_l$, where $l = 1,2,\ldots,m$.

The simultaneous confidence intervals have the following form

$$\hat{\gamma}_l \pm k_\alpha S.E.(\hat{\gamma}_l)$$

and the following properties

$$\boldsymbol{P}\left(\left|\frac{\widehat{\boldsymbol{\gamma}}_{\boldsymbol{l}}-\boldsymbol{\gamma}_{\boldsymbol{l}}}{\boldsymbol{S.E.}(\widehat{\boldsymbol{\gamma}}_{\boldsymbol{l}})}\right| \leq \boldsymbol{k}_{\boldsymbol{\alpha}}\right) = \boldsymbol{1}-\boldsymbol{\alpha} \text{ for } l = 1,2,\ldots,m.$$

Scheffe's Method: In this method, we would like to find a number k_α such that

$$\boldsymbol{P}\left(\left|\frac{\hat{\gamma}-\gamma}{S.E.(\hat{\gamma})}\right| \leq k_\alpha\right) = \mathbf{1} - \boldsymbol{\alpha} \text{ for } \textbf{all} \text{ contrasts}$$

It turns out that

$$k_\alpha = \sqrt{(k-1)F_{\alpha,k-1,n-k}}$$

Then we can calculate as many as confidence intervals as we wish, each of them have the following form:

$$\hat{\gamma}_l \pm k_\alpha S.E.(\hat{\gamma}_l)$$

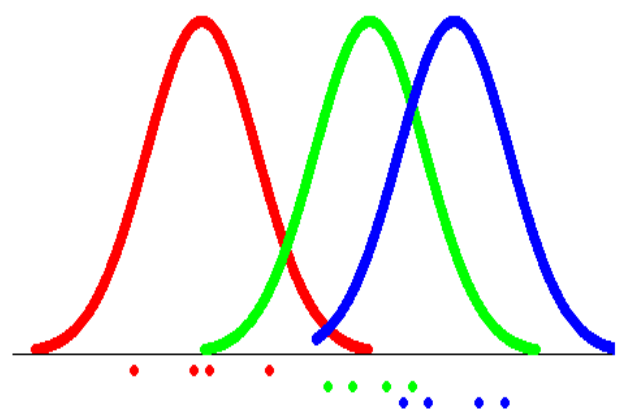
We reject the null hypothesis H_0: $\gamma = 0$ versus H_0: $\gamma \neq 0$ if

$$|\hat{\gamma}_l| > k_\alpha \; S.E.(\hat{\gamma}_l) = ME = B$$

Example 1 (Cont.): Suppose we would like to compare three brands of gasoline.

Case # 2

Brand 1	Brand 2	Brand 3
15.2	18.5	19.6
16.1	17.5	19.3
16.8	18.2	18.4
15.9	17.8	18.7
$\bar{y}_1 = 16$	$\bar{y}_2 = 18$	$\bar{y}_3 = 19$
$s_1 = 0.66$	$s_2 = 0.44$	$s_3 = 0.55$



Analysis of Variance Table

Response: y

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
f	2	18.667	9.3333	30.216	0.0001016 ***
Residuals	9	2.780	0.3089		

Example 1 (Cont.): Find a 95% simultaneous confidence intervals for $\mu_1 - \mu_2, \mu_1 - \mu_3$ and $\mu_2 - \mu_3$.

$$S.E.(\bar{y}_{i.} - \bar{y}_{j.}) = \sqrt{MS_{Error}}\sqrt{\frac{1}{n_i} + \frac{1}{n_j}} \quad \text{for } i \neq j,\ i,j = 1,2,3$$

$$= \sqrt{0.3089}\sqrt{\frac{1}{4} + \frac{1}{4}} = 0.393$$

$$k_\alpha = \sqrt{(k-1)F_{\alpha,k-1,n-k}} = \sqrt{(3-1)F_{0.05,2,9}} = \sqrt{(3-1)4.26} = 2.919$$

Therefore 95% simultaneous confidence intervals are:

$$\mu_1 - \mu_2 : \bar{y}_{1.} - \bar{y}_{2.} \pm 2.919(0.393) = 16 - 18 \pm 1.147 = -2 \pm 1.147$$

$$\mu_1 - \mu_3 : \bar{y}_{1.} - \bar{y}_{3.} \pm 2.919(0.393) = 16 - 19 \pm 1.147 = -3 \pm 1.146$$

$$\mu_2 - \mu_3 : \bar{y}_{2.} - \bar{y}_{3.} \pm 2.919(0.393) = 18 - 19 \pm 1.147 = -1 \pm 1.147$$

df_1

df_2

		1	2	3	4	5	6	7	8	9	10
8	0.25	1.54	1.66	1.67	1.66	1.66	1.65	1.64	1.64	1.63	1.63
8	0.1	3.46	3.11	2.92	2.81	2.73	2.67	2.62	2.59	2.56	2.54
8	0.05	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35
8	0.025	7.57	6.06	5.42	5.05	4.82	4.65	4.53	4.43	4.36	4.30
8	0.01	11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91	5.81
8	0.005	14.69	11.04	9.60	8.81	8.30	7.95	7.69	7.50	7.34	7.21
8	0.001	25.41	18.49	15.83	14.39	13.48	12.86	12.40	12.05	11.77	11.54
		1	2	3	4	5	6	7	8	9	10
9	0.25	1.51	1.62	1.63	1.63	1.62	1.61	1.60	1.60	1.59	1.59
9	0.1	3.36	3.01	2.81	2.69	2.61	2.55	2.51	2.47	2.44	2.42
9	0.05	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14
9	0.025	7.21	5.71	5.08	4.72	4.48	4.32	4.20	4.10	4.03	3.96
9	0.01	10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35	5.26
9	0.005	13.61	10.11	8.72	7.96	7.47	7.13	6.88	6.69	6.54	6.42
9	0.001	22.86	16.39	13.90	12.56	11.71	11.13	10.70	10.37	10.11	9.89

We can calculate confidence intervals for many other contrasts without affecting error bounds. On the other hand, Scheffe's intervals are quite **wide**.

Tukey's Method: If we know in advance that we are only interested in the **treatment differences**, this method can be used to construct simultaneous confidence intervals for $\mu_i - \mu_j$ for $i \neq j, i, j = 1,2, \dots, k$

In this method, we would like to find a number k_α such that

$$\boldsymbol{P}\left(\left|\frac{(\bar{y}_{i.}-\bar{y}_{j.})-(\mu_i-\mu_j)}{\boldsymbol{S.E.}(\bar{y}_{i.}-\bar{y}_{j.})}\right| \leq k_\alpha\right) = \mathbf{1} - \boldsymbol{\alpha}\ i \neq j,\ \ i, j = 1,2, \dots, k.$$

It turns out that

$$k_\alpha = \frac{q_\alpha}{\sqrt{2}}$$

where q_α can be found in R by qtukey$(1 - \alpha, k, df(Error) = n - k)$.

Example 1 (Cont.): Find a 95% simultaneous confidence intervals for $\mu_1 - \mu_2, \mu_1 - \mu_3$ and $\mu_2 - \mu_3$.

$$S.E.(\bar{y}_{i.} - \bar{y}_{j.}) = \sqrt{MS_{Error}}\sqrt{\frac{1}{n_i} + \frac{1}{n_j}} \quad \text{for } i \neq j,\ i,j = 1,2,3$$

$$= \sqrt{0.3089}\sqrt{\frac{1}{4} + \frac{1}{4}} = 0.393$$

$$k_\alpha = \frac{q_\alpha(0.95,3,9)}{\sqrt{2}} = \frac{3.948492}{\sqrt{2}} = 2.792$$

Therefore 95% simultaneous confidence intervals are:

$$\mu_1 - \mu_2 : \bar{y}_{1.} - \bar{y}_{2.} \pm 2.792(0.393) = 16 - 18 \pm 1.097 = -2 \pm 1.097$$

$$\mu_1 - \mu_3 : \bar{y}_{1.} - \bar{y}_{3.} \pm 2.792(0.393) = 16 - 19 \pm 1.097 = -3 \pm 1.097$$

$$\mu_2 - \mu_3 : \bar{y}_{2.} - \bar{y}_{3.} \pm 2.792(0.393) = 18 - 19 \pm 1.097 = -1 \pm 1.1097$$

There are other methods. For example, **Dunnett's** procedure can be used if one of the treatments is a control, and we only want to compare all others with control treatment (i.e. to see if any of them are better than a standard).

Bonferroni's Method: This method can be used to construct simultaneous confidence intervals for any contrasts. In this method we would like to find a number k_α such that

$$\boldsymbol{P}\left(\left|\frac{\hat{\gamma}_l-\gamma_l}{S.E.(\hat{\gamma}_l)}\right| \leq \boldsymbol{k_\alpha}\right) = \boldsymbol{1-\alpha} \text{ for } l = 1,2,\ldots,m.$$

It turns out that $k_\alpha = t_{\frac{\alpha}{2m},df=n-k}$.

Note: In Bonferroni's method, We find m individual $\left(1-\frac{\alpha}{m}\right)\%$ confidence intervals to obtain a $(1-\alpha)\%$ family-wise confidence intervals.

Example 1 (Cont.): Find a 94% simultaneous confidence intervals for $\mu_1 - \mu_2, \mu_1 - \mu_3$ and $\mu_2 - \mu_3$.

$$S.E.(\bar{y}_{i.} - \bar{y}_{j.}) = \sqrt{MS_{Error}}\sqrt{\frac{1}{n_i} + \frac{1}{n_j}} \quad \text{for } i \neq j,\ i,j = 1,2,3$$

$$= \sqrt{0.3089}\sqrt{\frac{1}{4} + \frac{1}{4}} = 0.393$$

$$k_\alpha == t_{\frac{\alpha}{2m}, df=n-k} = t_{\frac{0.06}{2(3)}, df=12-3} = t_{0.01,9} = 2.821$$

Therefore 94% simultaneous confidence intervals are:

$$\mu_1 - \mu_2 : \bar{y}_{1.} - \bar{y}_{2.} \pm 2.821(0.393) = 16 - 18 \pm 1.109 = -2 \pm 1.109$$

$$\mu_1 - \mu_3 : \bar{y}_{1.} - \bar{y}_{3.} \pm 2.821(0.393) = 16 - 19 \pm 1.109 = -3 \pm 1.109$$

$$\mu_2 - \mu_3 : \bar{y}_{2.} - \bar{y}_{3.} \pm 2.821(0.393) = 18 - 19 \pm 1.109 = -1 \pm 1.109$$

Cutoff probabilities

degrees of freedom	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
1	1.000	1.376	1.963	3.078	6.314	12.71	15.89	31.82	63.66	127.3	318.3	636.6
2	0.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.09	22.33	31.60
3	0.765	.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.21	12.92
4	0.741	.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	0.727	.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	0.718	.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	0.711	.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	0.706	.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	0.703	.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	0.700	.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	0.697	.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	0.695	.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	0.694	.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	0.692	.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	0.691	.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	0.690	.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	0.689	.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	0.688	.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.611	3.922
19	0.688	.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	0.687	.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850
21	0.663	.858	1.061	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	0.686	.858	1.061	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792

Example 1 (Cont.): Find a 95% simultaneous confidence intervals for $\mu_1 - \mu_2, \mu_1 - \mu_3$ and $\mu_2 - \mu_3$.

The margin of error (or error bound) for the confidence intervals are:

Scheffe's Method: B =ME = 1.147

Bonferroni's Method: B =ME = 1.109

Tukey's Method: B =ME = 1.097

R Code to Perform Multiple Comparisons Tests:

First use install.packages("multcomp") to install “multcomp” package.

```
>model2 = lm(y~f)
>library("multcomp")
>mtest = glht(model2,linfct=mcp(f="Tukey"))
```

You can write the contrast too.

```
>contrasts <- rbind("B1-B2"=c(1,-1,0), "B1-B3"=c(1,0,-1),
"B2-B3"=c(0,1,-1))

>mtest  <-glht(model2,linfct=mcp(f=contrasts))
```

R Code to Perform Tukey's Test s:

```
>mtest <- glht(model2,linfct=mcp(f="Tukey"))
>summary(mtest)

        Simultaneous Tests for General Linear Hypotheses
Multiple Comparisons of Means: Tukey Contrasts
Fit: aov(formula = y ~ f)
Linear Hypotheses:
                    Estimate Std. Error t value Pr(>|t|)
Brand 2 - Brand 1 == 0    2.000      0.393   5.089  0.00187 **
Brand 3 - Brand 1 == 0    3.000      0.393   7.634  < 0.001 ***
Brand 3 - Brand 2 == 0    1.000      0.393   2.545  0.07381 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
(Adjusted p values reported -- single-step method)
```

R Code to Perform Bonferroni's Tests:

```
>mtest  <-glht(model2,linfct=mcp(f=contrasts))
>summary(mtest,test=adjusted("bonferroni"))

              Simultaneous Tests for General Linear Hypotheses
Multiple Comparisons of Means: User-defined Contrasts
Fit: aov(formula = y ~ f)
Linear Hypotheses:
         Estimate Std. Error t value Pr(>|t|)
B1-B2 == 0   -2.000      0.393  -5.089  0.00196 **
B1-B3 == 0   -3.000      0.393  -7.634 9.64e-05 ***
B2-B3 == 0   -1.000      0.393  -2.545  0.09442 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
(Adjusted p values reported -- bonferroni method)
```

R Code to Construct Simultaneous Confidence Intervals:

```
>model2 = lm(y~f)
>mtest <- glht(model2,linfct=mcp(f="Tukey"))
>confint(mtest)

     Simultaneous Confidence Intervals
Multiple Comparisons of Means: Tukey Contrasts
Fit: aov(formula = y ~ f)
Quantile = 2.7907
95% family-wise confidence level
 Linear Hypotheses:
                      Estimate lwr      upr
Brand 2 - Brand 1 == 0  2.00000  0.90325  3.09675
Brand 3 - Brand 1 == 0  3.00000  1.90325  4.09675
Brand 3 - Brand 2 == 0  1.00000 -0.09675  2.09675
```

R Code to Construct Simultaneous Confidence Intervals:

```
>model2 = lm(y~f)
>TukeyHSD(model2, conf.level=0.95)

 Tukey multiple comparisons of means
   95% family-wise confidence level

Fit: aov(formula = y ~ f)

$f
                      diff        lwr        upr     p adj
Brand 2-Brand 1    2  0.90275798   3.097242   0.0016961
Brand 3-Brand 1    3  1.90275798   4.097242   0.0000852
Brand 3-Brand 2    1 -0.09724202   2.097242   0.0734796
```

What Have We Learned?

Analysis **o**f **va**riance (**ANOVA**) is used for comparing **several population means**.

Null hypothesis: H_0: $\mu_1 = \mu_2 = \mu_3 = \cdots = \mu_k$, for some $k \geq 2$.

Alternative hypothesis: H_a: **At least one** population mean is different from others. H_a: **At least two** population means are different.

Assumptions: The necessary assumptions are **independent simple random samples**, **equal variances** and **normal** populations.

Assumptions must be **checked** before proceeding any inference.

What Have We Learned (Cont.)?

Mean square error: MS_E is an estimate of population variance no matter if the null hypothesis is true or not.

$$MS_E = \mathrm{MS(W)} = \hat{\sigma}^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2 + \cdots + (n_k - 1)s_k^2}{N - k}$$

Test - statistic: $\mathrm{F} = \frac{\mathrm{MS}(Treatments)}{\mathrm{MS}(Error)} = \frac{SS(Treatments)/(k-1)}{SS(Error)/(n-k)}$

The null hypothesis is rejected if the test-statistic is sufficiently large.

We also learned about multiple comparisons tests such as **Bonferroni's method**, **Tukey's method** and **Scheffe's Method:.**